

28 April 2026

Exercise 1.

Select and Sorting

01 SELECT *

FROM employees

id	First-name	last-name	department	Salary	hire-date	City
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Pretoria
3	Cathy	Zulu	Finance	65000	2018-07-18	Cape Town
4	David	Mokoena	Marketing	50000	2021-11-05	Durban
5	Eva	Maidoo	IT	72000	2017-09-30	Johannesburg

02) SELECT DISTINCT department

FROM employees

department
IT
HR
finance
Marketing

03 SELECT First-name
last-name

FROM employees

ORDER BY Salary DESC

First-name	last-name
Eva	Maidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

04) SELECT id;

First-name;

last-name;

Salary

FROM employees

ORDER BY Salary DESC

Limit 3;

id	First-name	last-name	Salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

02 Filtering with WHERE

05) SELECT id;

First-name;

last-name;

department;

FROM employees

WHERE department = 'IT'.

id	First-name	last-name	department
5	Eva	Naidoo	IT
1	Alice	Green	IT

06) SELECT id, First-name

FROM employees

WHERE department = 'Finance'

AND salary > 60000

06) id	First-name	last-name	department	Salary
3	Cathy	Zulu	Finance	65000

07) SELECT id, first-name, last-name
 FROM employees
 WHERE department = 'HR'
 OR department = 'Marketing'

id	First-name	last-name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

08) SELECT id, first-name, last-name
 FROM employees
 WHERE department <> 'IT';

id	First-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

09) SELECT id, first-name, last-name, department
 FROM employees
 WHERE department IN ('IT', 'HR', 'Finance');

id	First-name	last-name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Maidoo	IT

10) SELECT id, first-name, last-name
FROM employees
WHERE department = 'IT'
AND Salary > 65000
AND City = 'Johannesburg';

id	First-name	last-name	department	Salary	City
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Maidoo	IT	72000	Not Jo'burg